Math 271 Mathematical Methods in Economics Tutorial 2 Part 1: Revision of Vector and Matrix Multiplication

1 Vector Multiplication

1.1 Two Ways to Multiply Vectors

For $u, v \in \mathbb{R}^m$:

Inner product (scalar):

$$u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{pmatrix}, \qquad v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix} \quad \Rightarrow \quad \langle u, v \rangle = \sum_{i=1}^{m} u_i \, v_i .$$

Outer product (matrix):

$$u \otimes v = \begin{pmatrix} u_1v_1 & u_1v_2 & \cdots & u_1v_m \\ u_2v_1 & u_2v_2 & \cdots & u_2v_m \\ \vdots & \vdots & \ddots & \vdots \\ u_mv_1 & u_mv_2 & \cdots & u_mv_m \end{pmatrix}$$

1.2 Pre-multiply a matrix with a row vector

Let $x = (\xi_1, \ldots, \xi_m)$ be a m-row vector and $A = (\alpha_{ij})$ be an $m \times n$ matrix.

$$(\xi_1 \ \xi_2 \ \ldots \ \xi_m) \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{pmatrix}$$

The row-vector x provides weights for linear combinations of the rows of A.

$$xA \ = \ \xi_1 \, A_1 + \xi_2 \, A_2 + \cdots + \xi_m \, A_m .$$

The product xA is a n-row vector. Its i-th component is the inner product of the row vector x with the i-th *col* of A:

$$(xA)_i \;=\; \sum_{i=1}^{n} \xi_r \, \alpha_{ri} \;=\; \langle x,\, A^i \rangle.$$

Example (3×3):

$$(\xi_1 \; \xi_2 \; \xi_3) \begin{pmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \end{pmatrix} (xA)_2 \;=\; \xi_1 \, \alpha_{12} + \xi_2 \, \alpha_{22} + \xi_3 \, \alpha_{32}$$

(so the 2nd entry is $\langle x, A_2 \rangle$)

1.3 Post-multiply a Matrix with a column vector

Let A be $m \times n$ and $y = (\eta_1, \ldots, \eta_n)^\top$.

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_n \end{pmatrix}$$

The column vector y provides weights for linear combinations of the *columns* of A.

$$Ay \;=\; \eta_1 \, A^1 + \eta_2 \, A^2 + \cdots + \eta_n \, A^n.$$

The product Ay is a m-column vector . Its i-th component equals the inner product of the i-th *row* of A with y:

$$(Ay)_i \;=\; \sum_{j=1}^{n} \alpha_{ij} \, \eta_j \;=\; \langle A_i,\, y \rangle.$$

Example (3×3, highlight one row):

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix}$$

$$(\mathrm{Ay})_2 \;=\; \alpha_{21} \, \eta_1 + \alpha_{22} \, \eta_2 + \alpha_{23} \, \eta_3.$$

(so the 2nd entry is $\langle A_2, y \rangle$)

2 Matrix $\times$ Matrix

2.1 Row Picture

Let A be $m \times n$, B be $n \times p$, so $C := AB$ is $m \times p$.

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} & \alpha_{14} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} & \alpha_{24} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} & \alpha_{34} \end{pmatrix} \begin{pmatrix} \beta_{11} & \beta_{12} & \beta_{13} & \beta_{14} & \beta_{15} \\ \beta_{21} & \beta_{22} & \beta_{23} & \beta_{24} & \beta_{25} \\ \beta_{31} & \beta_{32} & \beta_{33} & \beta_{34} & \beta_{35} \\ \beta_{41} & \beta_{42} & \beta_{43} & \beta_{44} & \beta_{45} \end{pmatrix}$$

Row wise linear-combination view: the i-th row of $C := AB$ is

$$C_i = \sum_{j=1}^{n} \alpha_{ij} B_j$$

i.e. the i-th row of AB is a linear combination of the *rows of post-multiplier* B with weights given by the entries of the i-th row of pre-multiplier matrix A .

2.2 Matrix $\times$ Matrix — Column picture

Let A be $m \times n$, B be $n \times p$, so $C := AB$ is $m \times p$.

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} & \alpha_{14} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} & \alpha_{24} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} & \alpha_{34} \end{pmatrix} \begin{pmatrix} \beta_{11} & \beta_{12} & \beta_{13} & \beta_{14} & \beta_{15} \\ \beta_{21} & \beta_{22} & \beta_{23} & \beta_{24} & \beta_{25} \\ \beta_{31} & \beta_{32} & \beta_{33} & \beta_{34} & \beta_{35} \\ \beta_{41} & \beta_{42} & \beta_{43} & \beta_{44} & \beta_{45} \end{pmatrix}$$

Column wise linear-combination view: the j-th col of $C := AB$ is

$$C^j = \sum_{j=1}^{n} \beta_{ij} A^j$$

i.e. the j-th column of AB is a linear combination of the *columns of pre-multiplier* A with weights given by the entries of the j-th column of post-multiplier matrix B .

2.3 Outer-product decomposition of AB - Col $\times$ Row

We can wite AB as a sum of outer products of **Columns of A and Rows of B**:

$$AB = \sum_{k=1}^{n} A_k \otimes B^k.$$

Here A_k is the k-th column of A (an $n \times 1$ vector) and B_k is the k-th row of B (a $1 \times p$ vector). Each outer product $A_k \otimes B_k$ produces an $n \times p$ matrix whose $(i, j)^{th}$-entry equals $a_{ik} b_{kj}$.

2.4 Inner-product decomposition of AB - Row $\times$ Col

Standard Class XII stuff - refer $NCERT$ if you don't remember!